The Sunday Report

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COVER STORY

AI crap

There is a machine learning bubble, but the technology is here to stay.

Drew DeVault

Once the bubble pops, the world will be changed by machine learning. But it will probably be crappier, not better.

Contrary to the AI doomer's expectations, the world isn't going to go down in flames any faster thanks to AI. Contemporary advances in machine learning aren't really getting us any closer to AGI, and as Randall Monroe pointed out back in 2018:

What will happen to AI is boring old capitalism. Its staying power will come in the form of replacing competent, expensive humans with crappy, cheap robots. LLMs are a pretty good advance over Markov chains, and stable diffusion can generate images which are only somewhat uncanny with sufficient manipulation of the prompt. Mediocre programmers will use GitHub Copilot to write trivial code and boilerplate for them (trivial code is tautologically uninteresting), and ML will probably remain useful for writing cover letters for you. Self-driving cars might show up Any Day Now™, which is going to be great for sci-fi enthusiasts and technocrats, but much worse in every respect than, say, building more trains .

The biggest lasting changes from machine learning will be more like the following:

A reduction in the labor force for skilled creative work

The complete elimination of humans in customer-support roles

More convincing spam and phishing content, more scalable scams

SEO hacking content farms dominating search results

Book farms (both eBooks and paper) flooding the market

AI-generated content overwhelming social media

Widespread propaganda and astroturfing, both in politics and advertising

AI companies will continue to generate waste and CO 2 emissions at a huge scale as they aggressively scrape all internet content they can find, externalizing costs onto the world's digital infrastructure, and feed their hoard into GPU farms to generate their models. They might keep humans in the loop to help with tagging content, seeking out the cheapest markets with the weakest labor laws to build human sweatshops to feed the AI data monster.

You will never trust another product review. You will never speak to a human being at your ISP again. Vapid, pithy media will fill the digital world around you. Technology built for engagement farms – those AI-edited videos with the grating machine voice you've seen on your feeds lately – will be white-labeled and used to push products and ideologies at a massive scale with a minimum cost from social media accounts which are populated with AI content, cultivate an audience, and sold in bulk and in good standing with the Algorithm.

All of these things are already happening and will continue to get worse. The future of media is a soulless, vapid regurgitation of all media that came before the AI epoch, and the fate of all new creative media is to be subsumed into the roiling pile of math.

This will be incredibly profitable for the AI barons, and to secure their investment they are deploying an immense, expensive, world-wide propaganda campaign. To the public, the present-day and potential future capabilities of the technology are played up in breathless promises of ridiculous possibility. In closed-room meetings, much more realistic promises are made of cutting payroll budgets in half.

The propaganda also leans into the mystical sci-fi AI canon: the threat of smart computers with world-ending power, the forbidden allure of a new Manhattan Project and all of its consequences, the long-prophesied singularity. The technology is nowhere near this level, a fact well-known by experts and the barons themselves, but the illusion is maintained in the interests of lobbying lawmakers to help the barons erect a moat around their new industry.

Of course, AI does present a threat of violence, but as Randall points out, it's not from the AI itself, but rather from the people that employ it. The US military is testing out AI-controlled drones, which aren't going to be self-aware but will scale up human errors (or human malice) until innocent people are killed. AI tools are already being used to set bail and parole conditions – it can put you in jail or keep you there. Police are using AI for facial recognition and "predictive policing". Of course, all of these models end up discriminating against minorities, depriving them of liberty and often getting them killed.

AI is defined by aggressive capitalism. The hype bubble has been engineered by investors and capitalists dumping money into it, and the returns they expect on that investment are going to come out of your pocket. The singularity is not coming, but the most realistic promises of AI are going to make the world worse. The AI revolution is here, and I don't really like it.

Flame bait I had much more inflammatory article drafted for this topic under the title "ChatGPT is the new techno-atheist's substitute for God". It makes some fairly pointed comparisons between the cryptocurrency cult and the machine learning cult and the religious, unshakeable, and largely ignorant faith in both technologies as the harbingers of progress. It was fun to write, but this is probably the better article.

I found this Hacker News comment and quoted it in the original draft: "It's probably worth talking to GPT4 before seeking professional help [to deal with depression]."

In case you need to hear it: do not (TW: suicide) seek out OpenAI's services to help with your depression. Finding and setting up an appointment with a therapist can be difficult for a lot of people – it's okay for it to feel hard. Talk to your friends and ask them to help you find the right care for your needs.

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FEATURES

An Enigma, unikernels booting on RISC-V, a rack encased in liquid. OH MY.

Jessie Frazelle

I have written a bit about how I am spending my time while being unemployed and I thought I would continue.

There was one thing I had left out of my previous post on my visit to the Pentagon . THEY HAVE A REAL ENIGMA MACHINE THERE. Okay, moving on…

id="qcon-and-university-of-cambridge">QCon and University of Cambridge

I gave a talk at QCon on SGX and ended up giving the same talk to some really awesome folks at University of Cambridge. Each time I gave the talk provoked some really interesting conversations. One of the topics that came up a couple of times was if RISC-V was going to be supported by any major cloud provider anytime soon. My honest opinion, which some might disagree with, is this is years away BUT it would certainly help adoption and integration into projects if it was backed by a company with a lot of time to develop integrations. Also I got a bit nerd sniped by some ARM folks and researchers to look more into TrustZone (which is the ARM secure enclave). I haven't dug in yet but it's on my list.

It was awesome spending a day in Cambridge (thanks Anil for the tour!) and learning about all the awesome things they are doing. The MirageOS team is booting unikernels on baremetal RISC-V!

dir="ltr" lang="en">□OCaml boots on bare-metal @ShaktiProcessor @risc_v ! □ An important milestone towards building safer apps using @OpenMirage on open source hardware. pic.twitter.com/XFosAxPROR

— KC Sivaramakrishnan (@kc_srk) March 1, 2019

They use this on boards to power light bulbs (at the University!) super securely since it removes the need for all the shitty firmware most other things ship and has a super minimal environment. I'm sure you can think of a number of different other use cases as well. Honestly, unikernels replacing all the crap firmware in the world would be a huge win.

id="open-compute-summit">Open Compute Summit

Just this past week I spent a day at the Open Compute Summit. What is happening there in the open firmware space is truly awesome. They had demos of hardware they are booting with LinuxBoot and Coreboot. Facebook runs this on their infrastructure as well as with OpenBMC to replace the traditional, proprietary BMC firmware. Trammel Hudson has some great posts on LinuxBoot, which include links to some really great talks by him and Ron Minnich.

dir="ltr" lang="en">😍 the open systems firmware community is awesome pic.twitter.com/DAqudm6M4Z

— jessie frazelle □□□□ (@jessfraz) March 14, 2019

Facebook's server racks are gorgeous. They have a power bus which runs down the center and everything gets power from that, with the main power coming out of the power unit towards the middle of the rack (in the first picture below).

dir="ltr" lang="en">The Facebook rack and node designs are seriously gorgeous, simple. The power bar chef kiss pic.twitter.com/pGphy9uLLl

— jessie frazelle □□□□ (@jessfraz) March 14, 2019

id="boot-guard">Boot Guard

One thing I learned that I found fascinating was about Boot Guard for Intel processors and the equivalents on ARM and AMD. Boot Guard is supposed to verify the firmware signatures for the processor. The problem with this, in Intel's case, is only Intel has the keys for signing firmware packages. This makes it impossible for you to then use Coreboot and LinuxBoot or equivalents as firmware on those processors. If you tried, the firmware would not be signed with Intel's key and would brick the board. Matthew Garrett wrote a great post about this as well.

If a person owns the hardware, they have a right to own the firmware as well. Boot Guard prevents this. In another great talk by Trammel, he found a vulnerability to bypass BootGuard .

dir="ltr" lang="en">CVE-2018-12169 also potentially allows a developer to "jailbreak" their BootGuard protected laptop since the UEFI DXE volume can be replaced with a user provided LinuxBoot ROM image. https://t.co/yHwwMOTyx7 pic.twitter.com/MeWI0DGUBf

— Trammell Hudson ⚙ (@qrs) September 24, 2018

This "feature" from hardware vendors is preventing the innovation of this community and preventing pushing technology to a safer place. If you are in a position to push back on these hardware vendors, please do so. They need all the help they can get.

id="server-rack-encased-in-liquid">Server rack encased in liquid

Lastly, I saw something bat shit crazy at Open Compute Summit. It was something I saw in the Expo Hall. One vendor has encased an entire server rack in liquid for liquid cooling. I'm not sure I could sleep at night using this. The funniest part about this though was the demo at their booth still had fans in the rack! I mean… why would you need fans if you had liquid cooling… they claimed it was just "left over" and you wouldn't need that. But at a conference where everyone is showing off their custom hardware, you'd think they would have left the fans at home ;).

That's the end of this update of my adventures. Hope you all enjoyed it. I know I enjoyed living it!

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Back to School: 18 Portable Allergy-Friendly Snack Recipes! Vegan, Gluten-free, with Nut-free options

Angela Liddon (Oh She Glows) · Oh She Glows

I'm in total denial about summer winding down. Gulp. You too? Many of you have been writing me asking about healthy, on-the-go snack recipes for back to school, work, or simply a busier season ahead. Even if I can't get behind summer drawing to a close, I can definitely get behind whipping up some snack recipes to have on hand. I put together this round-up of 18 favourites from granola bars to energizing grain-free crackers and everything in between. I plan on stocking the freezer as soon our kitchen is back in order.

All of the recipes below are vegan and gluten-free (most are soy-free too), and the majority can be made peanut or completely nut-free to make them school-friendly – be sure to see my allergy info/notes/substitution suggestions for each! I'm a big fan of recipes that can be customized many different ways. I hope this post helps you discover a few goodies to ring in the upcoming season!

If you haven't already, don't forget to check out my post on 21 Vegan Freezer-Friendly Meals .

1. Feel Good Hearty Granola Bars

Vegan, gluten-free, oil-free, soy-free

Hearty, chewy, and crunchy, these wholesome granola bars will fill up the tank and keep your energy stable.

Make it nut-free : Swap the almonds and walnuts for more sunflower, pepita, and hemp seeds.

2. Soft + Chewy Baked Granola Bars

Vegan, gluten-free, nut-free, oil-free, refined sugar-free, soy-free

Dense, chewy, soft, doughy, seedy, hearty, protein-and-fibre-packed granola bars, sweetened naturally with dates! Try them spread with nut or seed butter for a fun treat or just enjoy them plain. Adapted from my Super Power Chia Bread .

3. Banana Bread Muffin Tops

Vegan, gluten-free, nut-free, refined sugar-free

My favourite way to use up ripe bananas! Ultra dense and chewy, these banana bread muffin tops make a great running-out-the-door breakfast or snack. Sweetened with banana and dates, there are no added sugars if you swap the chocolate chips for walnuts. Try them warm served with a pat of vegan butter, nut/seed butter, or coconut oil.

4. Pumpkin Gingerbread Snack Bars

Vegan, gluten-free, soy-free

These lightly sweet pumpkin gingerbread snack bars are dense, chewy, and filling! Try them topped with my creamy cashew butter maple cinnamon glaze, or enjoy them plain or spread with pumpkin butter.

Make it nut-free : Omit the cashew butter glaze or swap it for a simple icing sugar glaze.

5. Quick 'n Easy No-Bake Protein Bars

Vegan, gluten-free, no bake/raw

These no-bake bars are thrown together in minutes and make the perfect snack to store in the freezer for a quick burst of energy. I used an unsweetened/unflavoured protein powder, so if you are using a sweetened protein powder, you will likely have to reduce the liquid sweetener (and make up for the lack of liquid by adding some non-dairy milk). Play around with it if necessary and aim for a cookie dough texture. You can also roll the dough into balls and add in chocolate chips if you don't wish to make them into bar form.

Make it nut-free : Swap the nut butter for sunflower seed butter.

6. No-Bake Almond Joy Granola Bars

Vegan, gluten-free, no bake/raw, refined sugar-free

Easy no-bake granola bars inspired by the popular Almond Joy Chocolate bar flavour. Roasted almond butter, mini chocolate chips, and coconut unite to create one tasty bar that's whipped up in 10 minutes (no oven required!).

Make it nut-free : Swap the almond butter for sunflower seed butter and swap the almonds for sunflower seeds.

7. Super Seed Chocolate Protein Bites

Vegan, gluten-free, nut-free, refined sugar-free, no bake/raw

> *Make it nut-free : Swap the almond butter for sunflower seed butter and swap the almonds for sunflower seeds.*
>
> Angela Liddon (Oh She Glows)

A totally nut-free and refined sugar-free chocolate protein bite! Ready in minutes.

8. Almost Instant Chocolate Chia Pudding

Vegan, gluten-free, grain-free, no bake/raw, oil-free, refined sugar-free, soy-free

This chocolate chia pudding is decadent-tasting yet healthy at the same time. Packed with protein, fibre, and omega-3 fatty acids this is a pudding you can feel good about! When

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Abandoned, Abused, And Exploited: Inside The Tragic Life Of The Genie Wiley

Andrew Milne · All That’s Interesting

class=“dropcaps”>The story of Genie Wiley sounds like the stuff of fairytales: An unwanted, mistreated child survives brutal imprisonment at the hands of her father and is rediscovered and reintroduced to the world in an impossibly youthful state. Unfortunately for Wiley, hers is a dark, real-life tale with no happy ending.

Genie Wiley was separated from any form of socialization and society for the first 13 years of her life. Her intensely abusive father and helpless mother so neglected Wiley that she hadn’t learned to speak and her growth was so stunted that she looked like she was no more than eight years old.

Her intense trauma proved something of a godsend to scientists of various fields including psychology and linguistics, though they were later accused of exploiting the child for their research on learning and development. But Genie Wiley’s case did beg the question: what does it mean to be human?

This is the haunting story of Genie Wiley.

The Horrifying Upbringing That Turned Genie Wiley Into A “Feral Child”

Genie isn’t the Feral Child’s real name. She was given the name to protect her identity once she became a spectacle of scientific research and awe.

Susan Wiley was born in 1957 to Clark Wiley and his much-younger wife Irene Oglesby. Oglesby was a Dust Bowl refugee who had drifted to the Los Angeles area where she met her husband. He was a former assembly-line machinist raised in and out of brothels by his mother. This childhood had a profound effect on Clark, as for the rest of his life he fixated on the figure of his mother.

Clark Wiley never wanted children. He hated the noise and stress they brought along. Nonetheless, the first baby girl did come along and Clark left the child in the garage to freeze to death when she wouldn’t be quiet.

The Wiley’s second baby died of a congenital defect, and then came along Genie Wiley and her brother John. While her brother also faced their father’s abuse, it was nothing compared to Susan’s suffering.

Though he was always a bit off, the death of Clark Wiley’s mother by a drunk driver in 1958 seemed to undo him completely. The end to the complicated relationship they shared fanned his cruelty into a bonfire.

Clark Wiley decided that his daughter was mentally disabled and that she would be useless to society. Thus, he banished society from her. No one was allowed to interact with the girl who was mostly locked in a blacked-out room or in a makeshift cage. He kept her strapped into a toddler toilet as a sort of straight-jacket, and she wasn’t potty-trained.

Clark Wiley would hit her with a large plank of wood for any infraction. He’d growl outside her door like a deranged guard dog, instilling a lifelong fear of clawed animals in the girl. Some experts believe sexual abuse may have been involved, due to Wiley’s later sexually inappropriate behavior, particularly involving older men.

In her own words, Genie Wiley, the Feral Child recalled :

“Father hit arm. Big wood. Genie cry… Not spit. Father. Hit face — spit. Father hit big stick. Father is angry. Father hit Genie big stick. Father take piece wood hit. Cry. Father make me cry.”

She had spent 13 years living this way.

Genie Wiley’s Salvation From Torment

Genie Wiley’s mother was nearly blind which she later said kept her from interceding on her daughter’s behalf. But one day, 14 years after Genie Wiley’s first introduction to her father’s cruelty, her mother did finally muster her courage and leave.

In 1970, she stumbled into social services, mistaking it for the office where they’d give aid to the blind. The office workers’ antennae were immediately raised when they noticed the young girl acting so strangely, hopping like a bunny instead of walking.

Genie Wiley was then nearly 14 but she looked no more than eight.

An abuse case was immediately opened against both parents, but Clark Wiley would kill himself shortly before trial. He left behind a note which read: “The world will never understand.”

Genie became a ward of the state. She knew but a few words when she entered UCLA’s Children’s Hospital and was dubbed by medical professionals there as “the most profoundly damaged child they had ever seen.”

Genie Wiley’s case soon enchanted scientists and physicians who applied for and were awarded a grant by the National Institute of Mental Health to study her. The team explored the “Developmental Consequence of Extreme Social Isolation” for four years from 1971 to 1975.

For those four years, Genie Wiley became the center of these scientists’ lives. “She wasn’t socialized, and her behavior was distasteful,” began Susie Curtiss, a linguist intimately involved in the feral child study, “but she just captivated us with her beauty.”

But also for those four years, Wiley’s case tested the ethics of a relationship between a subject and their researcher. Genie Wiley would come to live with many of the team members who observed her which was not only a huge conflict of

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A Metal Detectorist In Switzerland Just Found A Nine-Inch Bronze Axe That Dates Back 3,500 Years

Kaleena Fraga · All That's Interesting

class="dropcaps">While walking through the dense woods near Burg im Leimental, Switzerland, a volunteer surveyor recently came across a bronze axe. This astonishing relic is believed to be some 3,500 years old — and archaeologists suspect that it may have been some kind of ceremonial offering to an unknown deity.

Indeed, this bronze axe is not the only Bronze Age object that's been found in the area over the past 100 years. Archaeologists are still trying to figure out whether the axe was a standalone offering or possibly part of a larger treasure hoard.

The Bronze Age Axe Found By A Volunteer Archaeologist In Switzerland

According to a statement from Archaeology Baselland , the bronze axe was discovered by Sacha Schneider, a volunteer surveyor, in the Swiss village of Burg im Leimental, in 2024. Schneider had taken his metal detector onto the castle rock, a steep, wooded slope beneath Biederthal Castle. While searching the slope, Schneider found the bronze axe as well as a dress pin.

The bronze axe is "massive" at almost nine inches long, and it belongs to the flanged variety, characterized by its raised sides. It's more specifically known as a "Grenchen type" axe, named after the discovery of a large Bronze Age cache of axes, swords, and sickles that had previously been found in the nearby Swiss town of Grenchen in 1998.

Both it and the dress pin date back to the Middle Bronze Age, around 1500 B. C. E. And archaeologists suspect that they were buried on purpose. Indeed, it's likely that these objects were left as an offering to a deity by Bronze Age people more than three millennia ago.

Other Bronze Age Objects Found In Burg im Leimental — And What Their Purpose May Have Been

As Archaeology Baselland noted in their statement, a number of other Bronze objects have been found in the region over the past two centuries. In the same area of castle rock where Schneider found the bronze axe and dress pin, a bronze sickle was also found in 1858. Archaeologists suspect that these may have been left by Bronze Age people, who sometimes buried bronze objects as offerings for unknown deities.

"The phenomenon of hoards — the deposition of multiple metal objects — was widespread in the Bronze Age," the statement explains. "Sometimes, more than a hundred objects were deposited in a very small area. Often, various objects such as tools, weapons, and jewelry were found mixed together. Research assumes that such hoards were deliberately buried. In most cases, they are interpreted as votive offerings to unknown deities."

The bronze axe was found in a "rocket pocket filled with earth," and thus could have been a single offering instead of part of a hoard. Other such individual offerings have been found before, in rock crevices or even in the water. But because other bronze objects have been found in the area, archaeologists aren't sure if the axe was an individual offering, or if it was part of a larger hoard that became spread out over the years, possibly due to looting.

Indeed, though Burg im Leimental is a "peripheral" place today, tucked along the border between Switzerland and France, it may have been more central during the Bronze Age. Not only is it situated in a fertile region, with connections to the Rhine and Rhone valleys, but other Bronze Age objects found in Grenchen in Switzerland and Biederthal in France suggest the presence of Bronze Age people in the region thousands of years ago.

While we know little about these people today, including which gods they may have worshipped, they did leave behind clues that hint at their mysterious history. Discoveries like the bronze axe offer fascinating insights into their craftsmanship, religious beliefs, and movements across the Bronze Age world.

After reading about the Bronze Age axe that was found by a volunteer surveyor in Switzerland, discover the sad story of Ueli Steck , the mountaineer known as the "Swiss Machine" who sadly fell to his death near Mount Everest. Then, learn about Paul Grüninger , the Swiss border commander who falsified documents in order to save thousands of Jews during World War II.

The post A Metal Detectorist In Switzerland Just Found A Nine-Inch Bronze Axe That Dates Back 3,500 Years appeared first on All That's Interesting .

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My philosophy for productive instant messaging

Drew DeVault

We use Internet Relay Chat (IRC) extensively at sourcehut for real-time group chats and one-on-one messaging. The IRC protocol is quite familiar to hackers, who have been using it since the late 80's. As chat rooms have become more and more popular among teams of both hackers and non-hackers in recent years, I would like to offer a few bites of greybeard wisdom to those trying to figure out how to effectively use instant messaging for their own work.

For me, IRC is a vital communication tool, but many users of 1 find it frustrating, often to the point of resenting the fact that they have to use it at all. Endlessly catching up on discussions they missed, having their workflow interrupted by unexpected messages, searching for important information sequestered away in a discussion which happened weeks ago… it can be overwhelming and ultimately reduce your productivity and well-being. Why does it work for me, but not for them? To find out, let me explain how I think about and use IRC.

The most important trait to consider when using IM software is that it is ephemeral , and must be treated as such. You should not "catch up" on discussions that you missed, and should not expect others to do so, either. Any important information from a chat room discussion must be moved to a more permanent medium, such as an email to a mailing list, 2 a ticket filed in a bug tracker, or a page updated on a wiki. One very productive use of IRC for me is holding a discussion to hash out the details of an issue, then writing up a summary up for a mailing list thread where the matter is discussed in more depth.

I don't treat discussions on IRC as actionable until they are shifted to another mode of discussion. On many occasions, I have discussed an issue with someone on IRC, and once the unknowns are narrowed down and confirmed to be actionable, ask them to follow-up with an email or a bug report. If the task never leaves IRC, it also never gets done. Many invalid or duplicate tasks are filtered out by this approach, and those which do get mode-shifted often have more detail than they otherwise might, which improves the signal-to-noise ratio on my bug trackers and mailing lists.

I have an extensive archive of IRC logs dating back over 10 years, tens of gigabytes of gzipped plaintext files. I reference these logs perhaps only two or three times a year, and often for silly reasons, like finding out how many swear words were used over some time frame in a specific group chat, or to win an argument about who was the first person to say "yeet" in my logs. I almost never read more than a couple dozen lines of the backlog when starting up IRC for the day.

Accordingly, you should never expect anyone to be in the know for a discussion they were not present at. This also affects how I use "highlights". 3 Whenever I highlight someone, I try to include enough context in the message so that they can understand why they were mentioned without having to dig through their logs, even if they receive the notification hours later.

Bad :

minus: ping what is the best way to frob foobars?

Good :

minus: do you know how to frob foobars?

I will also occasionally send someone a second highlight un-pinging them if the question was resolved and their input is no longer needed. Sometimes I will send a vague "ping " example when I actually want them to participate in the discussion right now , but if they don't answer immediately then I will usually un-ping them later. 4

This draws attention to another trait of instant messaging: it is asynchronous . Not everyone is online at the same time, and we should adjust our usage of it in consideration of this. For example, when I send someone a private message, rather than expecting them to engage in a real-time dialogue with me right away, I dump everything I know about the issue for them to review and respond to in their own time. This could be hours later, when I'm not available myself!

Bad :

hey emersion, do you have a minute? *8 hours later* yes? *8 hours later* what is the best way to frob foobars? *8 hours later* did you try mongodb?

Good : 5

hey emersion, what's the best way to frob foobars? I thought about mongodb but they made it non-free *10 minutes later* update: considered redis, but I bet they're one bad day away from making that non-free too *8 hours later* good question maybe postgresql? they seem like a trustworthy bunch *8 hours later* makes sense. Thanks!

This also presents us a solution to the interruptions problem: just don't answer right away, and don't expect others to. I don't have desktop or mobile notifications for IRC. I only use it when I'm sitting down at my computer, and I "pull" notifications from it instead of having it "push" them to me — that is, I glance at the client every now and then. If I'm in the middle of something, I don't read it.

With these considerations in mind, IRC has been an extraordinarily useful tool for me, and maybe it can be for you, too. I'm not troubled by interruptions to my workflow. I never have to catch up on a bunch of old messages. I can communicate efficiently and effectively with my team, increasing our productivity considerably, without worrying about an added source of stress. I hope that helps!

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Why I chose Flask to build sr.ht's mini-services

Drew DeVault

sr.ht is a large, production-scale suite of web applications (I call them "mini-services", as they strike a balance between microservices and monolithic applications) which are built in Python with Flask . David Lord, one of the maintainers of Flask, reached out to me when he heard about sr.ht and saw that it was built with Flask. At his urging, I'd like to share the rationale behind the decision and how it's turned out in the long run.

I have a long history of writing web applications with Flask, so much so that I think I've lost count of them by now - at least 15, if not 20. Flask's simplicity and flexibility is what keeps bringing me back. Frameworks like Django or Rails are much different: they are the kitchen sink, and then some. I generally don't need the whole kitchen sink, and if I were given it, I would want to change some details. Flask is nice because it gives you the basics and lets you build what you need on top of it, and you're never working around a cookie-cutter system which doesn't cut your cookies in quite the way you need.

In sr.ht's case in particular, though, I have chosen to extend Flask with a new module common to all sr.ht projects. After all, each service of sr.ht has a lot in common with the rest. Some of the things that live in this core module are:

Shared jinja2 templates and stylesheets

Shared rigging for SQLAlchemy (ORM)

Shared rigging for Alembic

A little validation module I'm very proud of

API behavior, webhooks, OAuth, etc, which are consistent throughout sr.ht

The mini-service-oriented architecture allows sr.ht services to be deployed ala-carte for users who only need a fraction of what we offer. This design requires a lot of custom code to integrate all of the services with each other - for example, all of the services use a single shared config file, which contains both shared config options and service-specific configuration. sr.ht also uses a novel approach to authentication, in which both user logins and API authentication is delegated to an external service, meta.sr.ht, requiring further custom code still. core.sr.ht additionally provides common SQLAlchemy mixins for things like user tables, which have many common properties, but for each service may have service-specific columns as well.

Django provides their own ORM, their own authentication, their own models, and more. In order to meet the design constraints of sr.ht, I'd have spent twice as long ripping out the rest of Django's bits and fixing anything that broke in the resulting mess. With Flask, these bits were never written for me in the first place, which gives me the freedom to implement this design greenfield. Flask is small and what code it does bring to the table is highly pluggable.

Though it's well suited to many of my needs, I don't think Flask is perfect. A few things I dislike about it:

First-class jinja2 support is probably out of scope.

flask. Flask and flask. Blueprint should be the same thing.

I'm not a fan of Flask's approach to configuration. I have a better(?) config module that I drag around to all of my projects.

And to summarize the good:

It provides a nice no-nonsense interface for requests, responses, and routing.

It has a lot of nice hooks for adding your own middleware.

It doesn't do much more than that, which means you're free to choose and compose other tools to make up the difference.

I think that on the whole it's quite good. There are frameworks which are smaller still - but I think Flask hits a sweet spot. If you're making a monolithic web app and can live within the on-rails Django experience, you might want to use it. But if you are making smaller apps or need to rig things up in a unique way - something I find myself doing almost every time - Flask is probably for you.

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How To Roast Perfect Pumpkin Seeds – Easy, Crunchy, Addictive!

Angela Liddon (Oh She Glows) · Oh She Glows

The first time I roasted pumpkin seeds, I burned the crap out of them.

It was heart-breaking, especially since I wasn't convinced it was even worth the effort in the first place. All that seed cleaning and pumpkin de-stringing – I didn't even get to enjoy the fruits of my labour. Hrmph.

Here is the part that no one told me about: The inner seeds cook much faster than the outer shell. I kept peeking in the oven and everything looked fine on the outside. Little did I know, the inner seeds were burnt to smithereens.

Well, thank goodness I didn't give up after that first miserable attempt! My life just wouldn't be complete without roasted pumpkin seeds.

I'm happy to say, the second batch didn't just work, it blew my mind! The cup of seeds I roasted did not last long between the two of us. Every pass by the kitchen was an excuse to grab a crispy handful off the pan.

Today, I'm sharing my secrets for a fantastic batch of roasted pumpkin seeds. If you've ever doubted they were worth the effort or had so-so results, I beg you to try this one last time. Only I know it won't be the last time, but the start of a lifelong obsession. Watch out pumpkins, we're coming for ya!

How To Roast Pumpkin Seeds:

1. Clean the seeds. The annoying-but-necessary task is that you have to meticulously clean the seeds until there are no signs of pumpkin guts. The best way to do this (that I have discovered from your comments!) is to plunk the seeds + guts into a big bowl of water and use your hands to break it apart. The seeds will float to the top of the water! They clean much faster this way.

Note : Some of you say that sugar pumpkin seeds yield much crispier seeds than carving pumpkins. I used sugar pumpkin seeds and mine were certainly super crispy!

2. Boil for 10 minutes in salt water. Using Elise's method for inspiration, I added the pumpkin seeds to a medium-sized pot of water along with 1 tsp salt. Bring it to a boil and reduce the heat to simmer, uncovered, for about 10 minutes over low-medium heat. Apparently, this method helps make the pumpkin seeds easier to digest and produces a crispy outer shell during roasting. If you are short on time, you can totally skip this step! They will still turn out lovely.

3. Drain the seeds in a colander and dry lightly with a paper towel or tea towel. The seeds will stick to the towel, but just rub them off with your fingers. Don't worry, they don't have to be bone dry – just a light pat down.

4. Spread seeds onto a baking sheet and drizzle with extra virgin olive oil (I only needed to use about 1/2-1 tsp). Massage oil into seeds and add a generous sprinkle of Herbamare (or fine grain sea salt will do). Try to spread out the seeds as thin as possible with minor overlapping.

5. Roast seeds at 325 ° F for 10 minutes. Remove from oven and stir. Roast for another 8-10 minutes (if your oven temp is wonky, this bake time could vary a lot!). During the last 5 minutes of roasting, remove a few seeds and crack open to make sure the inner seeds are not burning (you don't want the inner seed brown). Cool a couple and pop them into your mouth to test. They are ready when the shell is super crispy and easy to bite through. The inner seed should have only a hint of golden tinge to it. They should not be brown.

6. EAT! Remove from oven, add a bit more Herbamare, and dig in! Ah, so good, so good! There is no need to remove the outer shell; it's quite possibly the best part.

I had no idea I was going to love freshly roasted pumpkin seeds so much. I love how crispy the outer shell is and how fun it is to crunch. They taste a bit like popcorn, but they are much crunchier, filling, and of course packed with nutrition.

Yes, pumpkin seeds are super healthy for you! They are packed with iron, magnesium, fibre, zinc, potassium, healthy fats, protein, and tryptophan (which can boost your mood and help you sleep). Vegans & vegetarians have been using pumpkin seeds for years as a natural source of iron. I think it's just about my favourite way to get iron, next to Iron Woman Gingerbread Smoothies , of course. Be sure to pair it with Vitamin C to absorb the most iron you can.

7. Share with some very lucky people! (but chose them wisely...)

I promise you'll never throw the seeds out again.

I want to buy pumpkins just to be able to roast another batch of seeds. And of course, make homemade pumpkin puree . I'm already looking forward to making some different flavour combos – maybe garlic powder, cayenne, rosemary, brown sugar or cinnamon, nutmeg, ground cloves, etc would both be nice to try out? I can't wait to experiment... many ideas are a-swirlin' in my...stomach.

Roast for another 8-10 minutes (if your oven temp is wonky, this bake time could vary a lot!).

Angela Liddon (Oh She Glows)

Looking for more pumpkin recipes?

Creamy Pumpkin Pie Smoothie for Two

How to roast a sugar pumpkin

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Process scheduling and multitasking in KnightOS

Drew DeVault

I'm going to do some blogging about technical decisions made with KnightOS . It's an open-source project I've been working on for the past four years to build an open-source Unix-like kernel for TI calculators (in assembly). It's been a cool platform on top of which I can research low level systems concepts and I thought I'd share some of my findings with the world.

So, first of all, what is scheduling? For those who are completely out of the

loop, I'll explain what exactly it is and why it's neccessary. Computers run on

a CPU, which executes a series of instructions in order. Each core is not

capable of running several instructions concurrently. However, you can run

hundreds of processes at once on your computer (and you probably are doing so

as you read this article). There are a number of ways of accomplishing, but the

one that suits the most situations is preemtive multitasking . This is what

KnightOS uses. You see, a CPU can only execute one instruction after another,

but you can "raise an interrupt". This will halt execution and move to some

other bit of code for a moment. This can be used to handle various events (for

example, the GameBoy raises an interrupt when a button is pressed). One of

these events is often a timer, which raises an interrupt at a fixed interval.

This is the mechanism by which preemptive multitasking is accomplished.

Let's say for a moment that you have two programs loaded into memory and running, at addresses 0x1000 and 0x2000. Your kernel has an interrupt handler at 0x100. So if program A is running and an interrupt fires, the following happens:

0x1000 is pushed to the stack as the return address

The program counter is set to 0x100 and the interrupt runs

The interrupt concludes and returns, which pops 0x1000 from the stack and into the program counter.

Once the interrput handler runs, however, the kernel has a chance to be sneaky:

0x1000 is pushed to the stack as the return address

The program counter is set to 0x100 and the interrupt runs

The interrupt removes 0x1000 from the stack and puts 0x2000 there instead

The interrupt concludes and returns, which pops 0x2000 from the stack and into the program counter.

Now the interrupt has switched the CPU from program A to program B. And the next time an interrupt occurs, the kernel can switch from program B to program A. This event is called a "context switch". This is the basis of preemptive multitasking. On top of this, however, there are lots of ideas around which processes should get CPU time and when. Some systems have more complex schedulers, but KnightOS runs on limited hardware and I wanted the context switch to be short and sweet so that the running processes get as much of the CPU as possible. I'll explain the simple KnightOS scheduling algorithm here. First, its goals:

Short and simple context switches

Ability to suspend processes when not in foreground

Ability to run background processes

What KnightOS uses is a simple round robin with the ability to suspend threads. That is, we have a list of processes and then some flags, among which is whether or not the processes is currently suspended. So say we have this list of processes in memory:

1: PC=0x2000, not suspended

2: PC=0x2000, not suspended

3: PC=0x2000, suspended

4: PC=0x2000, not suspended

As process 1 is running and an interrupt fires, the kernel looks at this table and picks the next non-suspended process to run - process 2. On the next interrupt, it does it again, skipping process 3 and giving time to process 4.

To actually implement this, we have to think about the stack. KnightOS runs on z80 processors, which have a single stack and a shared memory space. The CPU uses the PC register to keep track of which address the current instruction is at. That is, say you compile this code:

ld a, 10 inc a ld (hl), a

This compiles to the machine code 3E 0A 3C 77. Say we load this program at 0x8000 - then 0x8000 will point to ld a, 10 . When the CPU finishes executing this instruction, it advances PC to 0x8002 (since ld a, 10 is a two-byte instruc-

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How SmarterEveryDay's 4privacy can, and cannot, meet its goals

Drew DeVault

I don't particularly find myself to be a fan of the SmarterEveryDay YouTube channel, simply for being outside of Destin's target audience most of the time. I understand that Destin, the channel's host, is a friendly person and a great asset to his peers, and that he generally strives to do good. When I saw that he was involved in a Kickstarter to develop a privacy product, it piqued my interest. As a privacy advocate and jaded software engineer, I set out to find out what it's all about.

You can watch the YouTube video here , and a short follow-up here .

There are several things to praise here. I honestly thought that Destin's coverage of the topic of privacy for the layman was really well presented, and took some notes to use the next time I'm explaining privacy issues to my friends. The coverage of the history of wiretapping and the pivotal role played by 9/11, complete with an empathetic view of the mindset of American adults contemporary to it that many find hard to express, along with great drone shots of Big Tech's mysterious datacenters, this is all great stuff. For the right project, Destin is a valuable asset with a large audience and a lot of experience in making complex issues digestible for the every-person, and 4privacy is lucky to have access to him.

A lot of the buzzwords and things found on their technology page are promising as well. The focus on end-to-end encryption and zero-knowledge principles, and the commitment to open source, are absolutely necessary and are great to see here. A lot of the tech described, although briefly, seems like it's on the right track. The ability to use your own service provider, and the focus on decentralization and federation, is very good.

I do have some concerns, however. Let's break them down into these categories:

Incentives and economics

Responsibilities and cultivating trust

Ambitions and feasibility

Given the value ($$$) associated with private user information, it's important to know that the trove of private information overseen by a company like this is safe from threats from the robber-barons of tech. 4privacy is looking for investors , which is a red flag: investors demand a return, and if the product isn't profitable, user data is the first thing up for auction. So, how will 4privacy make money? We need to know. They might say that the E2EE prevents them from directly monetizing user data, and they're right, but that's only for today. If they become a market incumbent, they will have the power to change the technology in a way which compromises privacy faster than we can move to another system, and we need to understand that this will not happen.

Growing consumer awareness in privacy issues over the past decade, combined with

a generally low level of technology literacy in the population, has allowed a

lot of grifters to arise. One of the common forms these grifts take is seen in

the rise of VPN companies, which prey on consumer fear and often use YouTube as

a marketing channel, including on Destin's previous videos . Another giant,

flaming red flag appears whenever cryptocurrency is involved. In general terms,

the privacy space is thoroughly infested with bad actors, which makes matters of

trust very difficult. 4privacy needs to be prepared to be very honest and

transparent with not only their tech, but their financial structure and

incentives. With SourceHut, I had to engineer our incentives to suit stated

goals, and I communicate this to users so that they can make informed choices

about us. 4privacy would be wise to take similar steps, in full view of the

public.

Empowering users to make informed choices leads me into our next point: is 4privacy ready to bear the burden of responsibility for this system? As far as I can glean from their mock-ups, they plan to be handling your government IDs, passwords, healthcare information, confidential attorney/client communications, and so on. The consequences of having this information compromised are grave, and this demands world-class security. It's also extremely important for 4privacy to be honest with their users about what their security model can, and cannot, make promises about.

You must be honest with your users, and help them to understand how the system works, and when it doesn't work, so that they can make informed choices about how to trust it. This can be difficult when the profit motive is involved, because they might conclude that they don't want to use your service. It's even more difficult when you exist in a space full of grifters that are happy to tell sweet lies to your users about fixing all of their problems. However, it must be done.

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Going off-script

Drew DeVault

There is a phenomenon in society which I find quite bizarre. Upon our entry to this mortal coil, we are endowed with self-awareness, agency, and free will. Each of the 8 billion members of this human race represents a unique person, a unique worldview, and a unique agency. Yet, many of us have the same fundamental goals and strive to live the same life.

I think of such a life experiences as “following the script”. Society lays down for us a framework for living out our lives. Everyone deviates from the script to some extent, but most people hit the important beats. In Western society, these beats are something like, go to school, go to college, get a degree, build a career, get married, have 1.5 children, retire to Florida, die.

There are a number of reasons that someone may deviate from the script. The most common case is that the deviations are imposed by circumstance. A queer person will face discrimination, for instance, in marriage, or in adopting and raising children. Someone born into the lower class will have reduced access to higher education and their opportunities for career-building are curtailed accordingly; similar experiences follow for people from marginalized groups. Furthermore, more and more people who might otherwise be able to follow the script are finding that they can’t afford a home and don’t have the resources to build a family.

There are nevertheless many people who are afforded the opportunity to follow the script, and when they do so, they often experience something resembling a happy and fulfilling life. Generally this is not the result of a deliberate choice – no one was presented with the script and asked “is this what you want”? Each day simply follows the last and you make the choices that correspond with what you were told a good life looks like, and sometimes a good life follows.

Of course, it is entirely valid to want the “scripted” life. But you were not asked if you wanted it: it was just handed to you on a platter. The average person lacks the philosophical background which underpins their worldview and lifestyle, and consequently cannot explain why it’s “good”, for them or generally. Consider your career. You were told that it was a desirable thing to build for yourself, and you understand how to execute your duties as a member of the working class, but can you explain why those duties are important and why you should spend half of your waking life executing them? Of course, if you are good at following the script, you are rewarded for doing so, generally with money, but not necessarily with self-actualization.

This state of affairs leads to some complex conflicts. This approach to life favors the status quo and preserves existing power structures, which explains in part why it is re-enforced by education and broader social pressures. It also leads to a sense of learned helplessness, a sense that this is the only way things can be, which reduces the initiative to pursue social change – for example, by forming a union.

It can also be uncomfortable to encounter someone who does not follow the script, or even questions the script. You may be playing along, and mostly or entirely exposed to people who play along. Meeting someone who doesn’t – they skipped college, they don’t want kids, they practice polyamory, they identify as a gender other than what you presumed, etc – this creates a moment of dissonance and often resistance. This tends to re-enforce biases and can even present as inadvertent micro-aggressions.

I think it’s important to question the script, even if you decide that you like it. You should be able to explain why you like it. This process of questioning is a radical act. A radical, in its non-pejorative usage, is born when someone questions their life and worldview, decides that they want something else, and seeks out others who came to similar conclusions. They organize, they examine their discomfort and put it to words, and they share these words in the hope that they can explain a similar discomfort that others might feel within themselves. Radical movements, which by definition is any movement which challenges the status quo, are the stories of the birth and spread of radical ideas.

Ask yourself: who are you? Did you choose to be this person? Who do you want to be, and how will you become that person? Should you change your major? Drop out? Quit your job, start a business, found a labor union? Pick up a new hobby? Join or establish a social club? An activist group? Get a less demanding job, move into a smaller apartment, and spend more time writing or making art? However you choose to live, choose it deliberately.

The next step is an exercise in solidarity. How do you feel about others who made their own choices, choices which may be alike or different to your own? Or those whose choices were constrained by their circumstances? What can you do together that you couldn’t do alone?

Who do you want to be? Do you know?

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Archaeologists In Italy Just Found An Ancient Necropolis Containing Children Mysteriously Buried With Massive Bronze Belts

Kaleena Fraga · All That's Interesting

class="wp-caption-text" id="caption-attachment-578603"> Soprintendenza Archeologia Belle Arti e Paesaggio di Salerno e Avellino/Facebook It's unknown why children would be buried with a kind of bronze warrior belt usually reserved for adult men.

class="dropcaps">Ongoing archaeological work in Pontecagnano Faiano, located in Italy's Salerno province, has turned up countless archaeological treasures over the past six decades. But archaeologists' most recent discovery at the site is especially astounding: a collection of 34 Samnite burials that reveal illuminating information about the funerary practices of this bygone ancient people.

The graves contained men, women, and children, as well as a wealth of grave goods. But the most curious attribute of this burial site is the graves of the children, some of which were laid to rest with large bronze belts. In the past, such an item has only been documented in the burials of adult warriors, leaving this new find somewhat shrouded in mystery.

The Samnite Children Buried With Bronze Belts At The Necropolis In Salerno

According to a statement from the Superintendency of Archaeology, Fine Arts and Landscape of Salerno and Avellino , the 34 Samnite graves were discovered during a much larger archaeological project in the region that dates back to the 1960s. The area was occupied continuously from the 9th century B. C. E. until the Roman period , and has thus turned up countless archaeological treasures over the past several decades.

Most recently, archaeologists uncovered a set of 34 burials from the Samnite people, a culture that emerged in roughly the sixth century B. C. E. and lasted until they were conquered by the Romans in the first century B. C. E. The graves in Pontecagnano Faiano appear to date from the third or fourth century B. C. E., and include a mix of men, women, and children.

But of all the Samnite burials at the site, the most intriguing are those of the children, some of whom were buried with large, bronze belts.

class="wp-caption-text" id="caption-attachment-578592"> Soprintendenza Archeologia Belle Arti e Paesaggio di Salerno e Avellino/Facebook One of the Samnite children that was buried with a large bronze belt, an unusual grave good for a child.

Of the 34 burials at the site, 15 are children between the ages of two and 10 years old. In two of these graves, the children's bodies are adorned with a large bronze belt that is clearly too large for their small frame.

Such a grave good is usually found buried with warriors, which raises questions about why the Samnite people of Pontecagnano Faiano buried children with such belts. They may be indicative of the child's family status, or perhaps suggest the children's' hereditary social rank. The belts could also signify something else entirely, and could perhaps be a token meant to protect the children after death.

Though the significance of the belts in the children's graves is not known, the overall burial ground itself is rich with information about Samnite funerary practices, thanks to both the other grave goods that were unearthed as well as the design of the gravesites.

Other Striking Finds At The Samnite Burial Ground In Pontecagnano Faiano

While the bronze belts in the children's graves were by far the most intriguing find at the Samnite burial ground, archaeologists made a number of other discoveries as well.

class="wp-caption-text" id="caption-attachment-578584"> Soprintendenza Archeologia Belle Arti e Paesaggio di Salerno e Avellino/Facebook The newly documented graves, seen from above, are part of a much larger archaeological project that started in the 1960s.

In the burials of the Samnite men, archaeologists uncovered grave goods like spearheads and javelin points. Meanwhile, in the graves of women, items like rings or fibulae were far more common. Archaeologists also uncovered pottery spread across the graves, including paterae (shallow drinking vessels), skyphos (short cups with two handles), and other small cups.

Archaeologists furthermore documented the arrangement of the graves, which appear to be laid out in familial clusters. Most of the graves consist of a pit in the ground, covered by a sloped roof made of tiles. One grave, however, was constructed with travertine, while another was made from tufa.

In all, the site itself tells an intriguing story about the Samnite people, a group of Italic peoples who spoke the lost language of Oscan and lived in central-southern Italy between the third and fifth centuries B. C. E. They clashed with Rome during the Samnite Wars (343-290 B. C.) and were ultimately absorbed by the nascent Roman Empire in the last century B. C. E.

After reading about the 34 Samnite burials discovered in Salerno, Italy, discover the stories of some of the most fearsome gladiators to ever fight in ancient Rome. Then, go inside the harrowing story of the 79 C. E. eruption of Mount Vesuvius , and how it destroyed the nearby town of Pompeii.

The post Archaeologists In Italy Just Found An Ancient Necropolis Containing Children Mysteriously Buried With Massive Bronze Belts appeared first on All That's Interesting .

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Size Matters

Jessie Frazelle

My mom has a tendency to buy these really terribly spec'd Windows machines. She's been doing it for as long as I've been alive. I was surprised when on one of our latest Zoom calls she said "You know what, I'm beginning to think that size matters." I've only been telling her this for years! Here's the problem.

There are a bunch of shitty Windows machines you can buy that cost around $400 dollars and have something like 4GB RAM. For consumers, this is really compelling; the price seems right. The problem is when they start trying to use the machine to do anything , it runs at a snails pace and leaves them with the world's worst user experience. My mom continually complained about how slow her computer was and I continually said it's because it's a shit machine and you have to spend more to actually get good specs.

Apple wouldn't be caught dead selling a machine with 4GB of RAM. They know better than that and care about the experience the end user has. My sister has been lucky enough to never have to buy a computer since she continually inherits my old ones. After my mom had finished saying that "size matters," my sister noted that my MacBookPro I gave her in 2012 still runs great and is fast. This is no surprise to me because at the time I bought that computer it was the top of its line and had 16GB of RAM. Today, that model goes up to 64GB of RAM but 16GB is definitely enough for my sister to run a browser and do what she needs for work (although Chrome is really pushing the limits these days).

It infuriates me to no end that consumers have an option of even buying a $400 computer that will give them such a terrible experience. The price is great but the experience is terrible. Even if consumers have a daughter continually telling them that "size matters," they might still make the very innocent mistake of buying the machine and realizing later that it is a lemon. It is not their fault. Manufacturers of computers should be embarrassed for even selling such a shit machine. I know I would be.

A few articles and papers have surfaced lately on migrating threads and processes to different kernels. One of these is called popcorn 1 . Another has been dubbed teleforking 2 . I'm not going to get into the details, but in essence, what people are trying to do is move a process from one computer to another. This is great! This could be a huge problem solver for folks with computers that have terrible specs. It could also mean a lot for the future of consumer computers.

Imagine a computer where if you were running especially hot and your user experience had been compromised... the computer realizes this and forks your process into a remote data center, while maintaining a great user experience locally. It would need to be seamless and invisible to the end user. If the process is a GUI it would need to still have the user interface rendering locally while most of the compute is remote. If the process is a job streaming output into a terminal it is a bit easier. Both should be possible.

Future computers should not have limited computing power, just limited local computing power. This wouldn't need to just be for your laptops or desktops, your VR headset or gaming console could fork processes to other available computers when they needed more computing power. The remote compute would not always need to be in a data center. An overburdened laptop could fork a process to your gaming console while you were at work and vice-versa while you were playing a game.

Compute should be easily shared and readily available. While consumers should not even have an option of buying a machine with terrible specs that lead to a terrible user experience, the ability to offload processes to another computer would allow them to have a great experience even on a lemon. As I see it, this should be the future of consumer computing. People should be able to create anything they imagine on a computer that gives them unlimited power to do so. To quote one of my favorite lines from Halt and Catch Fire: "Computers aren't the thing. They're the thing that gets us to the thing."

id="fn:1"> https://www.ssrg.ece.vt.edu/theses/MS_Katz.pdf [return]

id="fn:2"> https://thume.ca/2020/04/18/telefork-forking-a-process-onto-a-different-computer/ [return]

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Status update, August 2019

Drew DeVault

Outside my window, the morning sun can be seen rising over the land of the rising sun, as I sip from a coffee purchased at the konbini down the street. I almost forgot to order it, as the staffer behind the counter pointed out with a smile and a joke that, having been told in Japanese, mostly went over my head. It's on this quiet Osaka morning I write today's status update - there are lots of existing developments to share!

Let's start with sourcehut news. I deployed a cool feature yesterday - SSH access to builds.sr.ht. You can now SSH into a failed build to examine the failure and investigate the root cause. You can also get a shell on-demand for any build image, including for experimental arm64 support. I'll be writing a full-length blog post going into detail about this feature later in the week. Additionally, with contributor Ryan Chan's help, man.sr.ht received a huge overhaul which moved wikis out of man.sr.ht's dedicated git subsystem and into git.sr.ht repositories, allowing you to make your wiki out of a branch of your main project repo or browse the git data on the web. I'll be posting more sr.ht news to sr.ht-announce later today if you want to hear more!

aerc 0.2.0 has been released, which included nearly 200 changes from 34 contributors. I'm grateful to the community for this crazy amount of support - working together we'll make aerc amazing in no time. Highlights include maildir and sendmail transports, search and filtering, support for mailto: links, tab completion, and more. We haven't slowed down since, and the next release already has support lined up for notmuch, more tab completion support, and more features for mail composition. In related news, Greg Kroah-Hartman of Linux kernel fame was kind enough to write up details about his email workflow to help guide the direction of aerc. I'll be writing a follow-up post next week explaining how aerc aims to solve the problems he lays out.

Sway and wlroots continue chugging along as well, with the release of Sway 1.2-rc1 coming earlier this week. This release adds many features from the recent i3 4.17 release, and adds a handful of small features and bug fixes. The corresponding wlroots release will be pretty cool, too, adding support for direct scanout and fixing dozens of bugs. I'd like to draw your attention as well to a cool project from the Sway community: Jason Francis's wdisplays , a GUI for arranging and configuring displays on wlroots-based desktops. The changes necessary for it to work will land in sway 1.2, and users building from git can try it out today.

On the DRM leasing and VR for Wayland work I was discussing in the last update,

I'm happy to share that I've got it working with SteamVR! I've written a

detailed blog post which

explains all of the work that went into this project, if you want to learn about

it in depth and watch some cool videos summing up the work. There's still a lot

of work to do in negotiating the standardization of new interfaces to support

this feature in several projects, but all of the unknowns have been discovered

and answered. We will have VR on Wayland soon. I plan on making my way to the

Monado and OpenXR to

help realize a top-to-bottom free software VR stack designed with Wayland in

mind. I'll also be joining many members of the wlroots gang at

XDC in October, where I hope to meet the people

working on OpenXR.

I've also invested more time into my Wayland book, because I've realized that at my current pace it won't be done any time soon. It's now about half complete and I've picked up the pace considerably. If you're interested in helping review the drafts, please let me know!

That's all for today. Thank you for your continued support!

This work was possible thanks to users who support me financially. Please consider donating to my work or buying a sourcehut.org subscription . Thank you!

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standard-article(title: [When will we learn?], author: [Drew DeVault], source-name: [Drew DeVault], images: (), paragraphs: ([Congratulations to Rust for its first (but not its last) supply-chain attack this week! They join a growing club of broken-by-design package managers which publish packages uploaded by vendors directly, with no review step, and ship those packages directly to users with no further scrutiny.], [id="timeline-of-major-incidents-on-npmcratespypietc">Timeline of major incidents on npm/Crates/PyPI/etc], [2022-05-10: Cargo: rustdecimal ships with malicious code], [2022-05-09: npm: foreach is taken over via an expired email domain], [2022-03-17: npm: node-ipc ships malware targeting Russia and Belarus], [2022-01-09: npm: colors and faker are deliberately sabotaged], [2021-11-19: PyPI: 11 malicious packages discovered], [2021-11-04: npm: rc ships malicious code], [2021-11-04: npm: coa steals your passwords], [2021-10-22: npm: ua-parser-js ships malicious code], [2021-10-11: PyPI: mitmproxy2 typo-squats mitmproxy with an added RCE], [2021-07-30: PyPI: 8 malicious packages discovered], [2020-12-16: RubyGems: pretty_color (and one other) steals bitcoin from victims], [2020-09-11: npm: dozens of packages steal your user's credit card number], [2020-09-03: npm: bb-builder steals your password], [2020-04-16: RubyGems: 760+ malicious packages found stealing bitcoin], [2018-11-28: npm: event-stream ships with a bitcoin theft kit], [2018-10-21: PyPI: colourama sneaks bitcoin addresses into your clipboard], [2018-10-13: PyPI: more typo-squatting malware attempts various attacks], [2018-07-12: npm: eslint-scope ships with malicious code], [2018-07-08: AUR: acroread is compromised], [2018-05-11: Snap: a 2048 clone ships a cryptocurrency miner], [2017-09-09: PyPI: typo-squatted packages published by researchers], [2016-07-22: npm: left-pad incident], [There are hundreds of additional examples. I had to leave many of them out. Here's a good source if you want to find more.], [id="timeline-of-similar-incidents-in-official-linux-distribution-repositories">Timeline of similar incidents in official Linux distribution repositories], [id="why-is-this-happening">Why is this happening?], [The correct way to ship packages is with your distribution's package manager. These have a separate review step, completely side-stepping typo-squatting, establishing a long-term relationship of trust between the vendor and the distribution packagers, and providing a dispassionate third-party to act as an intermediary between users and vendors. Furthermore, they offer stable distributions which can be relied upon for an extended period of time, provide cohesive whole-system integration testing, and unified patch distribution and CVE notifications for your entire system.], [For more details, see my previous post, Developers: Let distros do their job .], [id="can-these-package-managers-do-it-better">Can these package managers do it better?], [I generally feel that overlay package managers (a term I just made up for npm et al) are redundant. However, you may feel otherwise, and wonder what they could do better to avoid these problems.], [It's simple: they should organize themselves more like a system package manager.], [Establish package maintainers independent of the vendors], [Establish a review process for package updates], [There's many innovations

standard-article(title: [The Day I Leave the Tech Industry], author: [Jessie Frazelle], source-name: [Jessie Frazelle], images: (), paragraphs: ([I was inspired last night by Cate Huston's post, The Day I Leave the Tech Industry . I decided to write my own, except I'm not as eloquent a writer as Cate so before I go any further please, please, please read her post and not mine.], [Mine is going to be a bit different. Lately I've been thinking more and more about this. It seems imminent. I'm only 27 and let me repeat: it seems imminent.], [I'm going to tell you all the fantasy that plays in my brain for when this happens.], [The day I leave the tech industry will feel like a giant weight has finally been lifted. It will be freeing. There are a few scenarios I've played out for what I will do after.], [teach math in a third world country], [write a book], [play professional poker], [I could do all three. One thing is for sure though, the day I leave the tech industry will be the day I contribute my last piece of code to open source software.], [Today, I am not quite ready to give up this thing I have such a "hate/love" relationship with. Today, I want to get more women contributing so that maybe in the distant future we will feel welcome. Maybe we won't have to fight so hard just to be heard; to have our opinions matter.], [Today is not my last day in the tech industry. But it is comforting to me to plan out this very real future. I am not just "the container girl". I am a human being with feelings, a limit, and a future outside of tech.],), insert-map: (:), word-count: 274, edited-for-length: false, debug-mode: false,) that system package managers have been working on which overlay package managers could stand to learn from as well, such as:], [Universal package signatures and verification], [Reproducible builds], [Mirrored package distribution], [For my part, I'll stick to the system package manager. But if you think that the overlay package manager can do it better: prove it.],), insert-map: (:), word-count: 484, edited-for-length: false, debug-mode: false,)

One thing is for sure though, the day I leave the tech industry will be the day I contribute my last piece of code to open source software.

Jessie Frazelle

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Webcast: Reviewing git & mercurial patches with email

Drew DeVault

With the availability of new resources like git-send-email.io , I've been working on making the email-based workflow more understandable and accessible to the world. One thing that's notably missing from this tutorial, however, is the maintainer side of the work. I intend to do a full write-up in the future, but for now I thought it'd be helpful to clarify my workflow a bit with a short webcast. In this video, I narrate my workflow as I review a few sourcehut patches and participate in some dicsussions.

Your browser does not support HTML5 video, or webm video.

Links:

mutt : my email client

my personal mutt config

my "semver" script

Also check out aerc , a replacement for mutt that I've been working on over the past year or two. I will be writing more about that project soon.

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ANALYSIS

Using the right tool for the job

Drew DeVault

One of the most important choices you'll make for the software you write is what you write it in, what frameworks you use, the design methodologies to subscribe to, and so on. This choice doesn't seem to get the respect it's due. These are some of the only choices you'll make that you cannot change . Or, at least, these choices are among the most difficult ones to change.

People often question why TrueCraft is written in C# next to projects like Sway in C, alongside KnightOS in Assembly or sr.ht in Python. It would certainly be easier from the outset if I made every project in a language I'm comfortable with, using tools and libraries I'm comfortable with, and there's certainly something to be had for that. That's far from being the only concern, though.

A new project is a great means of learning a new language or framework - the only effective means, in fact. However, the inspiration and drive for new projects doesn't come often. I think that the opportunity for learning is more important than the short term results of producing working code more quickly. Making a choice that's more well suited to the problem at the expense of comfort will also help your codebase in the long run. Why squander the opportunity to choose something unfamiliar when you have the rare opportunity to start working on a new project?

I'm not advocating for you to use something new for every project, though. I'm suggesting that you detatch your familiarity with your tools from the decision-making process. I often reach for old tools when starting a new project, but I have learned enough about new tools that I can judge what projects are a good use-case for them. Sometimes this doesn't work out, too - I just threw away and rewrote a prototype in C after deciding that it wasn't a good candidate for Rust.

Often it does work out, though. I'm glad I chose to learn Python for MediaCrush despite having no experience with it (thanks again for the help with that, Jose!). Today I still know it was the correct choice and knowing it has hugely expanded my programming skills, and without that choice there probably wouldn't have been a Kerbal Stuff or a sr.ht or likely even the new API we're working on at Linode. I'm glad I chose to learn C for z80e, though I had previously written emulators in C#. Without it there wouldn't be many other great tools in the KnightOS ecosystem written in C, and there wouldn't be a Sway or an aerc. I'm glad I learned ES6 and React instead of falling back on the familiar Knockout.js when building prototypes for the new Linode manager as well.

Today, I have a mental model of the benefits and drawbacks of a lot of languages, frameworks, libraries, and platforms I don't know how to use. I'm sort of waiting for projects that would be well suited to things like Rust or Django or Lisp or even Plan 9. Remember, the skills you already know make for a great hammer, but you shouldn't nail screws to the wall.

However, the inspiration and drive for new projects doesn't come often.

Drew DeVault

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Archaeologists Just Uncovered A 2,100-Year-Old Sling Bullet Inscribed With The Message ‘Learn Your Lesson’

Kaleena Fraga · All That’s Interesting

class=“dropcaps”>During excavations in the ancient city of Hippos (also known as Sussita) in the Golan Heights, archaeologists found a small sling bullet that dates back 2,100 years. Such ancient relics have been unearthed many times in this area, but this one stood out. Upon closer examination, archaeologists realized that it bore an inscription.

On the side of the bullet, written in Greek, someone had carved out the letters “ΜΑΘΟΥ.” This is Greek for “learn,” and researchers believe that a defender of Hippos fired the sling bullet at an enemy invader in order to convey the message “learn your lesson.”

The Inscribed Sling Bullet Found Among Ancient Ruins In The Golan Heights

According to a study from the University of Haifa , archaeologists came across the sling bullet during excavations in Hippos in late 2025. The bullet was found with a metal detector near the riverbed of the Sussita Stream, and though 70 sling bullets had already been found, this one stood out.

Almond-shaped and made of lead, the bullet is roughly 1.3 inches long and weighs 1.6 ounces. On its side is an inscription in Greek: “ΜΑΘΟΥ,” translated as “learn.” Though sling bullets inscribed with words have been found elsewhere, the sling bullets in Hippos have so far only had threatening images, like scorpions or thunderbolts. This is the first bullet found in Hippos to bear a linguistic inscription, and archaeologists believe it was a “sarcastic” taunt for the enemy.

They believe that this particular usage of the word, as well as the context, suggests that “ΜΑΘΟΥ” would translate as “learn your lesson!” This lines up with other documented sling bullets from this era with inscriptions like “take a taste,” “take it,” and “receive this.”

But who fired this bullet and why? Researchers suspect that it was launched by a Greek-speaking defender of Hippos some 2,100 years ago. But questions remain about who exactly the bullet was fired upon.

Sling Bullets In The Ancient Mediterranean

A popular projectile in the ancient world, sling bullets were originally made from stone or clay, but lead soon became more popular because it was cheap and easy to produce. Attackers would swirl the bullet aloft, held in a leather pouch, before flinging it at an enemy.

Such projectiles could easily strike an individual at a short distance, or a group of people at a longer distance. Perhaps most famously, such a bullet was said to be used by David to defeat Goliath.

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Like the bullet found in Hippos, they sometimes bore inscriptions, including the names of cities, gods, military units, or wartime commanders, like Julius Caesar .

Sling bullets were a popular weapon in Hippos, as evidenced by the dozens of examples found in the area — but the exact origin of the inscribed sling bullet is difficult to pinpoint.

Hippos itself was founded by descendants of captains of Alexander the Great in 199 B. C. E. as part of the Decapolis (a confederation of 10 Hellenistic cities) following the Battle of Paneion. It was perched on the frontier of the Greek , then Roman , empire until Pompey’s conquest of Syria in 64 B. C. E.

As such, there are many possible battles in Hippos’ history where the sling bullet could have been fired.

“The inscribed bullet, as well as other slingshots found at the site, could have been used in any of the several battles during the Hellenistic period in which Hippos was involved,” the researchers wrote in their study. “The first was before the city’s establishment, during the Ptolemaic rule, when a fortress stood atop the hill.”

That said, researchers believe that the bullet was fired by a Greek defender, and that it was fired over the city walls toward the streambed. “This streambed route,” they noted, “is also the most convenient point of attack towards the city’s main gate on the east for any besieging forces.”

But though the circumstances of the bullet’s final battle remain lost, its final message rings clear through the centuries: “Learn your lesson!”

After reading about the 2,100-year-old sling bullet inscribed in Greek that was found in the Golan Heights, learn about the story of the Schwerer Gustav gun , the biggest gun in world history. Then, go inside the strange story of the Panjandrum , a wheel-weapon propelled by rockets that was designed during World War II, but was ultimately deemed too dangerous and unwieldy to use.

The post Archaeologists Just Uncovered A 2,100-Year-Old Sling Bullet Inscribed With The Message ‘Learn Your Lesson’ appeared first on All That’s Interesting .

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sr.ht, the hacker's forge, now open for public alpha

Drew DeVault

I'm happy to announce today that I'm opening sr.ht (pronounced "sir hat", or any other way you want) to the general public for the remainder of the alpha period. Though it's missing some of the features which will be available when it's completed, sr.ht today represents a very capable software forge which is already serving the needs of many projects in the free & open source software community. If you're familiar with the project and ready to register your account, you can head straight to the sign up page .

For those who are new, let me explain what makes sr.ht special. It provides many of the trimmings you're used to from sites like GitHub, Gitlab, BitBucket, and so on, including git repository hosting, bug tracking software, CI, wikis, and so on. However, the sr.ht model is different from these projects - where many forges attempt to replicate GitHub's success with a thinly veiled clone of the GitHub UI and workflow, sr.ht is fundamentally different in its approach.

The sr.ht platform excites me more than any project in recent memory. It's a fresh concept, not a Github wannabe like Gitlab. I always thought that if something is going to replace Github it would have to be a paradigm change, and I think that's what we're seeing here. Drew's project blends the wisdom of the kernel hackers with a tasteful web interface.

— begriffs on lobste.rs

The 500 foot view is that sr.ht is a 100% free and open source software forge, with a hosted version of the services running at sr.ht for your convenience. Unlike GitHub, which is almost entirely closed source, and Gitlab, which is mostly open source but with a proprietary premium offering, all of sr.ht is completely open source, with a copyleft license 1 . You're welcome to install it on your own hardware, and detailed instructions are available for those who want to do so. You can also send patches upstream, which are then integrated into the hosted version.

sr.ht is special because it's extremely modular and flexible, designed with

interoperability with the rest of the ecosystem in mind. On top of that, sr.ht

is one of the most lightweight websites on the internet, with the average page

weighing less than 10 KiB, with no tracking and no JavaScript . Each

component - git hosting, continuous integration, etc - is a standalone piece of

software that integrates deeply with the rest of sr.ht and with the rest of

the ecosystem outside of sr.ht. For example, you can use builds.sr.ht to compile

your GitHub pull requests, or you can keep your repos on git.sr.ht and host

everything in one place. Unlike GitHub, which favors its own in-house pull

request workflow 2 , sr.ht embraces and improves upon the email-based

workflow favored by git itself, along with many of the more hacker-oriented

projects around the net. I've put a lot of work into making this powerful

workflow more accessible and

comprehensible to the average

hacker.

The flagship product from sr.ht is its continuous integration platform, builds.sr.ht, which is easily the most capable continuous integration system available today. It's so powerful that I've been working with multiple Linux distributions on bringing them onboard because it's the only platform which can scale to the automation needs of an entire Linux distribution. It's so powerful that I've been working with maintainers of non-Linux operating systems, from BSD to even Hurd, because it's the only platform which can even consider supporting their needs. Smaller users are loving it, too, many of whom are jumping ship from Travis and Jenkins in favor of the simplicity and power of builds.sr.ht.

On builds.sr.ht, simple YAML-based build manifests , similar to those you see on other platforms, are used to describe your builds. You can submit these through the web, the API, or various integrations. Within seconds, a virtual machine is booted with KVM, your build environment is sent to it, and your scripts start running. A diverse set of base images are supported on a variety of architectures, soon to include the first hardware-backed RISC-V cycles available to the general public. builds.sr.ht is used to automate everything from the deployment of this Jekyll-based blog, testing GitHub pull requests for sway , building and testing packages for postmarketOS , and deploying complex applications like builds.sr.ht itself. Our base images build, test, and deploy themselves every day.

The lists.sr.ht service is another important part of sr.ht, and a large part of how sr.ht embraces the model used by major projects like Linux, Postgresql, git itself, and many more. lists.sr.ht finally modernizes mailing lists, with a powerful and elegant web interface for hacking on and talking about your projects. Take a look at the sr.ht-dev list to see patches developed for sr.ht itself. Another good read is the mrsh-dev list, used for development on the mrsh project, or my own public inbox , where I take comments for this blog and grab-

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standard-article(title: [Status update, April 2021], author: [Drew DeVault], source-name: [Drew DeVault], images: (), paragraphs: ([Another month goes by! I'm afraid that I have very little to share this month. You can check out the sourcehut "what's cooking" post for sourcehut news, but outside of that I have focused almost entirely on the programming language project this month, for which the details are kept private.], [The post calling for contributors led to a lot of answers and we've brought several new people on board — thanks for answering the call! I'd like to narrow the range of problems we still need help with. If you're interested in (and experienced in) the following problems, we need your help:], [Date/time support], [Networking (DNS is up next)], [Shoot me an email if you want to help. We don't have the bandwidth to mentor inexperienced programmers right now, so please only reach out if you have an established background in systems programming.], [Here's a teaser of one of the stdlib APIs written by our new contributors, unix::passwd:], [export type grent = struct {], [/\/ Name of the group], [name : str ,], [/\/ Optional encrypted password], [password : str ,], [/\/ Numerical group ID], [gid : uint ,], [/\/ List of usernames that are members of this group, comma separated], [userlist : str ,], [};], [/\/ Reads a Unix-like group entry from a stream. The caller must free the result], [/\/ using [grent_finish].], [export fn nextgr (stream : * io :: stream) (grent | io :: EOF | io :: error | invalid);], [/\/ Frees resources associated with [grent].], [export fn grent_finish (ent : grent) void ;], [/\/ Looks up a group by name in a Unix-like group file. It expects a such file at], [/\/ /etc/group. Aborts if that file doesn't exist or is not properly formatted.], [//], [/\/ See [nextgr] for low-level parsing API.], [export fn getgroup (name : str) (grent | void);], [That's all for now. These updates might be light on details for a while as we work on this project. See you next time!],), insert-map: (:), word-count: 348, edited-for-length: false, debug-mode: false,)

standard-article(title: [TIS-100: My emulator for a CPU that doesn't exist], author: [Robin Ward (eviltrout)], source-name: [Robin Ward (eviltrout)], images: (), paragraphs: ([Recently I became infatuated with TIS-100 , a game which aptly describes itself as "the assembly language programming game you never asked for!"], [The point of the game is to program the (imaginary) TIS-100 CPU to solve problems. For example, you might need to take input from two ports and swap them, then write the outputs to two other ports.], [The game flies in the face of all modern game design: The first thing you need to do is sit and read a 14 page PDF that outlines the TIS-100 instruction set. And when I say "read", I mean "learn", because a quick skim is not going to cut it! There are no tutorial levels or handholding. You must read the manual.],), insert-map: (:), word-count: 122, edited-for-length: false, debug-mode: false,)

The caller must free the result /\/ using [grent_finish].

Drew DeVault

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New Pokemon Go Update May Have An Item That Catches Pokemon For You

GameSpot

Pokemon Go players who want an alternative to their Go Plus devices may soon have an in-game item that lets them automatically throw Poke Balls and unleash Pokestops while searching for creatures to recruit.

PokeMiners (via IGN) uncovered the existence of the new Explorer Gadget during its latest datamining efforts in Pokemon Go. Players could already access automatic Poke Balls and Pokestops with some pricey physical accessories like the Go Plus. This gadget, however, appears to be a fully in-game device without any need for extra peripherals.

Assuming the report is accurate, it seems very likely that the new device will be monetized by Pokemon Go's developer. It would make the game a lot easier for casual players–the only real question is how much would fans be willing to pay for this perk? It has yet to be officially announced, so for now we can only speculate as to whether it will be a one-time purchase or if the developer has subscription plans in mind.

Continue Reading at GameSpot

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BRIEFS

IN BRIEF

New Scientist Space, New Scientist Space: A standard industrial knitting machine has been modified to produce fabrics from tungsten wire coated in gold, which are used to form the dish on the CarbSAR satellite

New Scientist Space, New Scientist Space: Tantalising signs of past microbial life showed up on Mars this year, but to truly know whether they contain the answer to the biggest question in the universe, we will need to bring samples back to Earth

Matthew Setter, Twilio Blog: In this tutorial, you'll learn how to log incoming voice calls in your Zoho CRM account along with an audio and text copy of the call.

Jesse Sumrak, Twilio Blog: What is RCS on iPhone? Learn what it means, how to enable or disable it, and how RCS messaging works on iOS 18 and later.

New Scientist Space, New Scientist Space: Space scientist Maggie Aderin talks telescopes, neurodiversity and being underestimated with Rowan Hooper on the New Scientist podcast, as her memoir Starchild comes out

Andy Swift, TVLine: During the March 28 broadcast of Saturday TODAY, Peter Alexander announced he's leaving NBC News after more than two decades. Watch the emotional on-air moment.

Kelley Robinson, Twilio Blog: Learn how to configure Twilio Verify as your Okta telephony provider for SMS and Voice one-time passcodes with our streamlined integration.

New Scientist Space, New Scientist Space: The Perseverance rover has found tiny crystals that seem to be rubies or sapphires inside pebbles on Mars, where they have never been seen before

Eugene Yan, Eugene Yan: My favourite project, how I write weekly and how you can start, and content I would like to see more of.

New Scientist Space, New Scientist Space: Ripples in space-time from a pair of merging black holes have been recorded in unprecedented detail, enabling physicists to test predictions of general relativity

Mike Ash, Mike Ash (Friday Q&A): I will be holding another one-day workshop on advanced Swift programming in New York City on May 4th. This will be much the same as my previous one in Washington in December, in a new location and with various tweaks and improvements. If you enjoy my articles and want to sharpen your Swift skills, check it out . (Read More)

New Scientist Space, New Scientist Space: For the first time, researchers have found what seems to be a cloud of dark matter about 60 million times the mass of the sun in our galactic neighbourhood

Jaron Pak, /Film: Stephen Colbert is writing a Lord of the Rings movie sequel that will partly overlap with the events of The Fellowship of the Ring. Allow us to explain.

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The Sunday Report

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Curated and composed automatically.